HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed a **Python** application integrating **LLM APIs** with customer data, enabling production-ready AI features for hospitality, retail, and leisure sectors.
- Designed and implemented prompt engineering strategies and tool-calling mechanisms to structure LLM outputs for downstream processing, improving data accuracy by **18%**.
- Built **Python** data pipelines to ingest and process diverse customer datasets for AI model training and inference, handling approximately **3,500 events** per second during peak hours.
- Containerized the AI application using **Docker** and deployed it to a Kubernetes cluster, ensuring scalable and resilient operation on **AWS**.
- Collaborated with product managers and data scientists to define technical requirements and influence the AI product roadmap, contributing to the launch of new generative AI capabilities.
- Wrote comprehensive unit and integration tests for **Python** services, achieving over **85% code coverage** and reducing post-deployment bugs by **22%**.
- Contributed to the design and implementation of a **RAG (Retrieval-Augmented Generation)** system, integrating a vector database for efficient knowledge retrieval to enhance LLM responses.

SKILLS

Languages – Python, SQL, R

AI/ML – LLMs, Prompt Engineering, Tool Calling, Structured Outputs, RAG, Generative AI, scikit-learn, pandas, NumPy

Databases – SQL, PostgreSQL, MySQL, Oracle, Vector Databases

Cloud & DevOps – AWS, Docker, Kubernetes, Jenkins, Git, CI/CD

Frameworks & APIs – Spring Boot, FastAPI, Flask, REST APIs, LangChain

PROJECTS

Customer Data Integration for AI Feature Generation

- Engineered Python data pipelines to extract, transform, and load customer data from various sources, preparing it for use in AI applications.
- Implemented robust error handling and logging mechanisms, reducing data processing failures by **15%** and improving overall pipeline stability.
- Designed and built REST APIs using **FastAPI** to serve processed data to downstream AI model inference services.

LLM-Powered Knowledge Retrieval System

- Developed a **RAG** system that integrates a vector database with LLMs to provide contextually relevant answers from a large corpus of documents.
- Implemented prompt engineering techniques to optimize LLM interactions, resulting in a **20% improvement** in response relevance and accuracy.
- Containerized the application using **Docker** for consistent deployment across development and production environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)